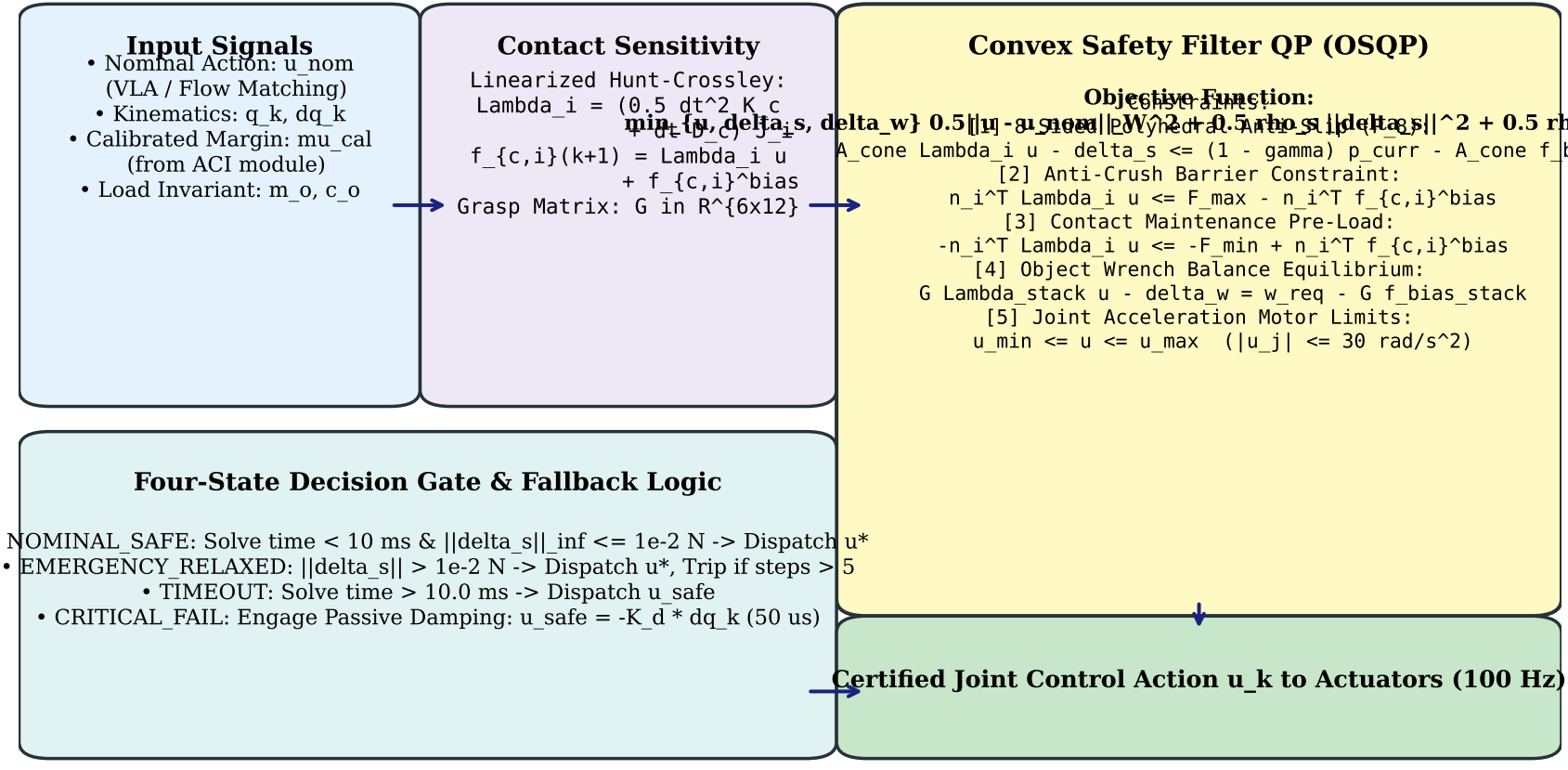

Input Signals
• Nominal Action: u_nom (VLA / Flow Matching)
• Kinematics: q_k, dq_k
• Calibrated Margin: mu_cal (from ACI module)
• Load Invariant: m_o, c_o
Contact Sensitivity
Linearized Hunt-Crossley:
Lambda_i = (0.5 dt^2 K_c + dt D_c) J_i
f_{c,i}(k+1) = Lambda_i u + f_{c,i}^bias
Grasp Matrix: G in R^{6x12}
Convex Safety Filter QP (OSQP)
Objective Function:
min_{u, delta_s, delta_w} 0.5||u - u_nom||_W^2 + 0.5 rho_s ||delta_s||^2 + 0.5 rho_w ||delta_w||^2
Constraints:
[1] Shifted Polyhedral Anti-Slip (FCA):
A_cone Lambda_i u - delta_s <= (1 - gamma) p_curr - A_cone f_bias
[2] Anti-Crush Barrier Constraint:
n_i^T Lambda_i u <= F_max - n_i^T f_{c,i}^bias
[3] Contact Maintenance Pre-Load:
-n_i^T Lambda_i u <= -F_min + n_i^T f_{c,i}^bias
[4] Object Wrench Balance Equilibrium:
G Lambda_stack u - delta_w = w_req - G f_bias_stack
[5] Joint Acceleration Motor Limits:
u_min <= u <= u_max (|u_j| <= 30 rad/s^2)
Four-State Decision Gate & Fallback Logic
• NOMINAL_SAFE: Solve time < 10 ms & ||delta_s||_inf <= 1e-2 N -> Dispatch u*
• EMERGENCY_RELAXED: ||delta_s|| > 1e-2 N -> Dispatch u*, Trip if steps > 5
• TIMEOUT: Solve time > 10.0 ms -> Dispatch u_safe
• CRITICAL_FAIL: Engage Passive Damping: u_safe = -K_d * dq_k (50 us)
Certified Joint Control Action u_k to Actuators (100 Hz)